

Phelix Estinvil

Multidisciplinary Technician & Technical Problem-Solver

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Professional Summary

Methodical, hands-on technical professional who learns new systems quickly, isolates faults systematically, and carries work from diagnosis through implementation and testing. Experience combines field service, fabrication, customer communication, live operations, and independent technical development.

Experience

Marine Mechanic - Northside Marina

Sesuit Harbor · East Dennis, Massachusetts

- **Diesel and gasoline engine diagnostics:** Diagnose no-start conditions and operating faults in diesel and gasoline marine engines. Isolate mechanical, electrical, and fuel-system causes using engine symptoms, diagnostic findings, and service information.
- **Diagnostic trouble-code analysis:** Interpret engine diagnostic trouble codes (DTCs) and correlate them with operating symptoms and diagnostic test results to identify underlying faults and determine the required repair.
- **Fuel delivery and water contamination:** Identify water-contaminated fuel and troubleshoot fuel-delivery faults affecting engine starting and operation. Distinguish fuel-related problems from electrical and mechanical faults to direct corrective work.
- **Starting, charging, and electrical systems:** Use multimeters, diagnostic equipment, and service information to trace faults in engine-starting and charging circuits. Identify wiring and component failures affecting reliable engine operation.
- **Trim and vessel-control systems:** Troubleshoot trim and vessel-control systems, isolating circuit and component faults that prevent correct operation and determining the required repairs.

Owner / Remodeler - Independent Remodeling

Approximately six years

- **Carpentry and installation:** Performed residential carpentry, layout, installation, repair, and finish work. Applied those skills to renovation projects from initial planning through final delivery.
- **Estimates and material planning:** Estimated project requirements and organized materials, tools, and supplies. Matched preparation to the scope of work and the planned sequence of the job.
- **Scheduling and jobsite execution:** Managed project schedules and day-to-day jobsite work as an independent business owner. Coordinated the work needed to move each renovation toward completion.
- **Jobsite problem-solving:** Assessed unexpected structural or installation problems encountered during remodeling. Adjusted the approach to address existing conditions and complete the required work.
- **Customer communication and quality control:** Discussed project expectations and progress with homeowners throughout the job. Reviewed completed work for quality and handled final delivery.

Experience

Founder / DJ / Event Technician - DJ Phelix

Cape Cod & South Shore

- **Client consultation and event planning:** Discuss the occasion, venue, audience, music preferences, and schedule with clients. Use those details to plan entertainment for weddings, private parties, corporate functions, and public events.
- **Music preparation:** Organize music around the event's theme and audience, including decade sets, nightlife programming, and family-friendly selections. Prepare a musical direction with room for requests and changes during the event.
- **Live mixing and audience response:** Select and mix tracks in real time, adjusting song choice, transitions, and energy to the people in the room. Shift between background music and dance-floor entertainment as the event progresses.
- **Sound-system setup and operation:** Configure microphones, playback equipment, mixers, speakers, and signal routing for each event. Manage audio levels and system settings throughout the performance.
- **MC and microphone operation:** Deliver announcements and introductions and coordinate music around speeches and scheduled moments. Manage microphone use alongside the event's music and entertainment.
- **Lighting setup and operation:** Set up and operate event lighting and lighting controllers to support the venue's atmosphere. Adjust lighting to suit the music and different stages of the event.
- **Live technical troubleshooting:** Trace audio, cabling, power, wireless, and equipment problems during live events. Work through connections, signal paths, and settings to identify the fault and restore operation.
- **Karaoke hosting:** Cue participant song selections and operate microphones alongside backing tracks. Coordinate performances and music to keep the session moving and guests involved.
- **Equipment logistics and venue coordination:** Organize equipment transport, setup, and breakdown for each venue. Coordinate timing and technical requirements with clients, venue staff, performers, and other vendors.
- **Booking communication and promotion:** Handle inquiries and communicate event details with clients and venues. Maintain DJ Phelix's website, upcoming-event information, and promotional materials to support bookings and audience outreach.

Technical Capabilities

Marine & mechanical: Diesel and gasoline diagnostics; fuel, starting, charging, trim, and vessel-control systems; multimeter testing; preventive maintenance.

Electronics & embedded: ESP32 / ESP32-S3; LVGL; PCB prototyping; sensors and battery monitoring; CAN bus; RS-485 / DMX512; Wi-Fi, BLE, and ESP-NOW.

Software & fabrication: Python; C/C++; Git / GitHub; Linux and Windows; PlatformIO, Arduino, ESP-IDF, and KiCad; technical documentation; 3D printing; carpentry and remodeling.

Selected Technical Projects

M5Stack DJ Smart Hub

Distributed lighting control

Distributed ESP32-S3 lighting-control system combining live audio analysis, scene logic, low-latency wireless transport, interface controls, and DMX output. Develop LVGL touchscreen interfaces and diagnostic, configuration, firmware-update, and field-service concepts; maintain source code, architecture documentation, and revision history in Git.

ESP32 Smart-Light Retrofit

Embedded hardware & fixture integration

Rechargeable RGB-fixture retrofit with internal DMX injection, wired fallback, ESP-NOW control, fixture profiles, and safety states. Evaluate control electronics, communications, USB-C power, battery management, and LED-control architectures for distributed fixtures.

PhelixSlicer

Software development & build workflows

Orca-based slicer fork focused on Flashforge Adventurer 3 Pro support, reproducible Windows builds, and production profile validation.

Embedded Electronics & Power-System Development

Independent technical projects

Prototype ESP32-based monitoring and control systems using voltage, current, and sensor measurements. Work with ADCs, shunts, battery systems, CAN transceivers, motor drivers, and RS-485 interfaces; troubleshoot hardware/software integration and perform bench diagnostics using multimeters and laboratory power supplies.

Computer & Server Systems

Development infrastructure & troubleshooting

Configure Windows and Linux systems for development, networking, storage, and remote access. Work with Git workflows, local servers, SSH, and computer hardware; diagnose GPU, network, storage, driver, and operating-system problems.

Education & Training

Cape Cod Regional Technical High School

Carpentry Program · Harwich, Massachusetts

Vocational training in carpentry, construction practices, tools, measurements, jobsite procedures, and residential building methods.

Project portfolio: phelixestinvil.com/projects/

DJ Phelix: djphelix.com | Clever Cat Company: clevercatcompany.com